

Math

Bellat Sir

☐ Single valued function: A function $w = f(z)$ is called a single valued in a domain S if and only value of w corresponds to each value of z in S .

If $w = f(z) = z^2 + 2$ then w is called a single valued function of z since to each value of z there is only one value of w .

☐ Multiple valued function:

more than one

$$z = f(z) = z^{\frac{1}{2}} = \sqrt{z} = \pm 2$$

☐ Real and Imaginary part of complex variable $z = x + iy$

Let $w = f(z) = u + iv$ be a single valued function of $z = x + iy$

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Now responding $x+iy$ for z we have $u+iv$

$$u+iv = f(x+iy)$$

Then equating real and imaginary parts we have

$$u = u(x, y) \text{ and } v = v(x, y)$$

$$\text{If } w = z^2 \text{ then } u+iv = e^{x+iy}$$

$$= e^x \cdot e^{iy}$$

$$= e^x (\cos y + i \sin y)$$

$$\boxed{e^{i\theta} = \cos \theta + i \sin \theta}$$

$$u = e^x \cos y = u(x, y) \text{ and } v = e^x \sin y = v(x, y)$$

□ Analytic Function

Differential coefficient in a region or domain

Let $f(z)$ be a single valued function defined

in a region or domain R , Then the derivative

of $f(z)$ is defined as $f'(z) = \lim_{\Delta z \rightarrow 0} \frac{f(z+\Delta z) - f(z)}{\Delta z}$

Then $f(z)$ is called Analytic Function
differentiable at z .



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Analytic function : A single valued function $f(z)$ is said to be analytic in Region R if the derivative $f'(z)$ exists.

25-Nov-24

Lecture-3

Cauchy - Riemann Equations :

Necessary and sufficient condition for the function $f(z) = u(x, y) + iv(x, y)$ to be analytic

① Necessary condition for $f(z)$ to be analytic if $f(z) = u(x, y) + iv(x, y)$ is analytic in a region R then it is necessary for the four partial derivatives $u_x = \frac{\partial u}{\partial x}$ $v_x = \frac{\partial v}{\partial x}$

$u_y = \frac{\partial u}{\partial y}$ $v_y = \frac{\partial v}{\partial y}$ should exist and

satisfy the Cauchy Riemann equation

$$u_x = v_y, \quad u_y = -v_x$$

② Necessary & sufficient condition for the $f(z)$ to be analytic :

The continuous single valued function

$f(z) = u(x, y) + iv(x, y)$ is analytic in a region R if the four partial derivatives



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u_x, v_x, u, v_y exist, are continuous and satisfy the Cauchy-Riemann equation
 $u_x = v_y, v_x = -u_y$ in R .

Cauchy-Riemann equation in polar form:

$$\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}, \quad \frac{\partial v}{\partial r} = -\frac{1}{r} \frac{\partial u}{\partial \theta}$$

verify the condition Cauchy-Riemann eqn satisfied for the following functions.

(i) $f(z) = e^{z^2}$

(ii) $f(z) = \cos 2z$

(i) we have $f(z) = u + iv = e^{(x+iy)^2} = e^{x^2 - y^2 + 2ixy} = e^{x^2 - y^2} \cdot e^{2ixy}$

$$u = e^{x^2 - y^2} \cos 2xy$$

$$v = e^{x^2 - y^2} \sin 2xy$$

$$u + iv = e^{x^2 - y^2} (\cos 2xy + i \sin 2xy)$$

$$u = e^{x^2 - y^2} \cos 2xy$$

$$v = e^{x^2 - y^2} \sin 2xy$$

$$u_x = 2x e^{x^2 - y^2} \cos 2xy$$

$$\frac{\partial u}{\partial x} = 2x e^{x^2 - y^2} \cos 2xy$$

$$\frac{\partial u}{\partial x} = 2x e^{x^2 - y^2} \cos 2xy$$

$$\frac{\partial v}{\partial y} = -2y e^{x^2 - y^2} \cos 2xy$$



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Lecture 4

Laplace eqn: In two dimension a second order partial differential equation $\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0$ or

$\nabla^2 \phi = 0$ is called Laplace eqn. The operator ∇^2 is called the Laplacian where $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$

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v.v.1

Harmonic Function: A function Laplace eqn satisfy is called Harmonic. It is not non-Harmonic.

Conjugate Harmonic Function: If $f(z) = u + iv$ is an analytic function, then u and v are called conjugate harmonic functions.

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

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$$\frac{\partial^2 \psi}{\partial x^2} + \frac{1}{r} \frac{\partial \psi}{\partial r} + \frac{1}{r^2} \frac{\partial^2 \psi}{\partial \theta^2} = 0$$

$\psi = r^2$



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Example

Find the harmonic conjugate of the function

$$u = x^3 + 6x^2y - 3xy^2 - 2y^3$$

Answer: we have $\frac{\partial u}{\partial x} = 3x^2 + 12xy - 3y^2$

$$\frac{\partial^2 u}{\partial x^2} = 6x + 12y$$

আবার

$$\frac{\partial u}{\partial y} = 6x^2 - 6xy - 6y^2$$

$$\frac{\partial^2 u}{\partial y^2} = -6x - 12y$$

$$\text{Now, } \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 6x + 12y - 6x - 12y$$

$$= 0$$

u is harmonic Function. Science u is harmonic.

Therefore it will be possible to Find the harmonic conjugate v of u . Now by Cauchy

Riemann eqn. we

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad -\frac{\partial u}{\partial y} = \frac{\partial v}{\partial x}$$

Now,

$$\frac{\partial v}{\partial y} = 3x^2 + 12xy - 3y^2$$

$$v = 3x^2y + 6xy^2 - y^3 + F(x)$$

$$\frac{\partial v}{\partial x} = 6xy + 6y^2 + F'(x)$$



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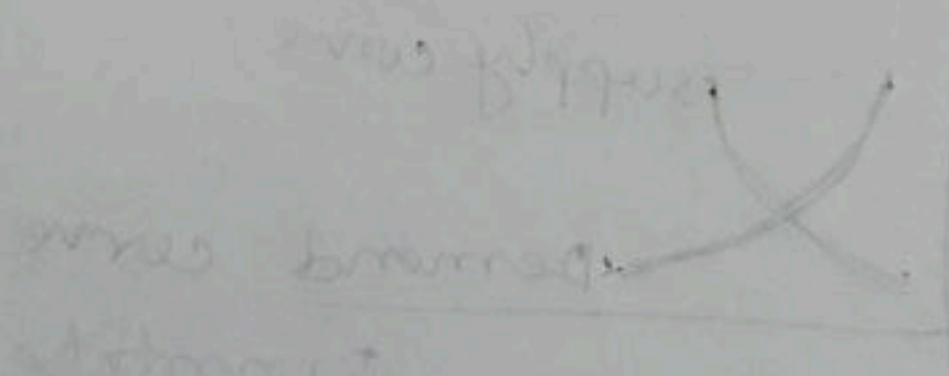
$$6xy + 6y^2 + F'(x) = -6x^2 + 6xy + 6y^2$$

$$F'(x) = -6x^2 + \cancel{6xy} + \cancel{6y^2} - \cancel{6xy} - \cancel{6y^2}$$
$$= -6x^2$$

$$F(x) = -2x^3 + c$$

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09-12-20

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M. construction of a analytic function:

if $f(z) = u(x, y) + iv(x, y)$ is analytic and

if $\frac{du}{dx} = \phi_1(x, y)$

$\frac{du}{dy} = \phi_2(x, y)$ and $\frac{dv}{dx} = \psi(x, y) = 0$

$\frac{dv}{dy} = \psi_2(x, y)$ then

$f'(z) = \phi_1(z, 0) - i\phi_2(z, 0)$

Milne Thomson method

$f(z)$ (বা কণ্ডে রচা)

$f'(z)$ (integration করে) $F(z)$ পাওয়া।

$f(z) = \int [\phi_1(z, 0) - i\phi_2(z, 0)] dz + c$

$f'(z) = \phi_1(z, 0) + i\psi_2(z, 0)$

এর integration করে $f(z)$

পাওয়া যায়

Plan Title :

Q1 show that, $u = 3x^2y + 2x^2 - y^3 - 2y^2$ is harmonic and hence find its harmonic conjugate v if $f(z) = u + iv$ is analytic.

we have $u = 3x^2y + 2x^2 - y^3 - 2y^2$

$$\frac{\partial u}{\partial x} = 6xy + 4x$$

$$= \phi_1(x, y)$$

$$\frac{\partial^2 u}{\partial x^2} = 6y + 4$$

$$\frac{\partial v}{\partial y} = 3x^2 - 3y^2 - 4y = \phi_2(x, y)$$

$$\frac{\partial^2 v}{\partial y^2} = 0 - 6y - 4$$

Now, $\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 6y + 4 - 6y - 4 = 0$

$$f'(z) = \phi_1(z, 0) - i \phi_2(z, 0)$$

$$= 4z - 3iz^2$$

$$f(z) = 2z^2 - iz^3$$

$$f(z) = u + iv = 2(x+iy)^2 - i(x+iy)^3$$

$$f(z) = u + iv = 2(x^2 + 2ixy - y^2) - i(x^3 + 3ix^2y - 3xy^2 - iy^3) \\ = 2(x^2 - y^2) + 3x^2y - y^3 + i(4xy - x^3 + 3xy^2 + y^3)$$

$$v = 4xy - x^3 + 3xy^2 + C$$

Question :

show that

$$u = 2x^2 - x^3 + 3xy^2$$

is harmonic

same answer:

Question :

show that

$$u = e^{x^2 - y^2} \cos 2xy$$

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Math

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Cauchy's Integral Formula:

Let $f(z)$ be analytic inside and on a simple closed curve c . If a is any point inside c . Then $f(a) = \frac{1}{2\pi i} \oint_c \frac{f(z)}{z-a} dz$, where c is the function counter clockwise direction.



EXAMPLE-182

show that, $\frac{1}{2\pi i} \oint_c \frac{e^z}{z-2} dz = \begin{cases} e^2 & \text{if } c \text{ is the circle } |z|=3 \\ 0 & \text{if } c \text{ is the circle } |z|=2 \end{cases}$

If $f(z)$ is the analytic inside and on the boundary c of a simply connected closed region R , Then

$$\frac{1}{2\pi i} \oint_c \left\{ \frac{1}{z-a} \pm \frac{1}{z-b} \right\} f(z) dz = \begin{cases} f(a) \pm f(b) & \text{if the point } a \text{ and } b \\ & \text{are inside } c. \\ f(a) & \text{if only point } a \text{ is inside } c. \end{cases}$$

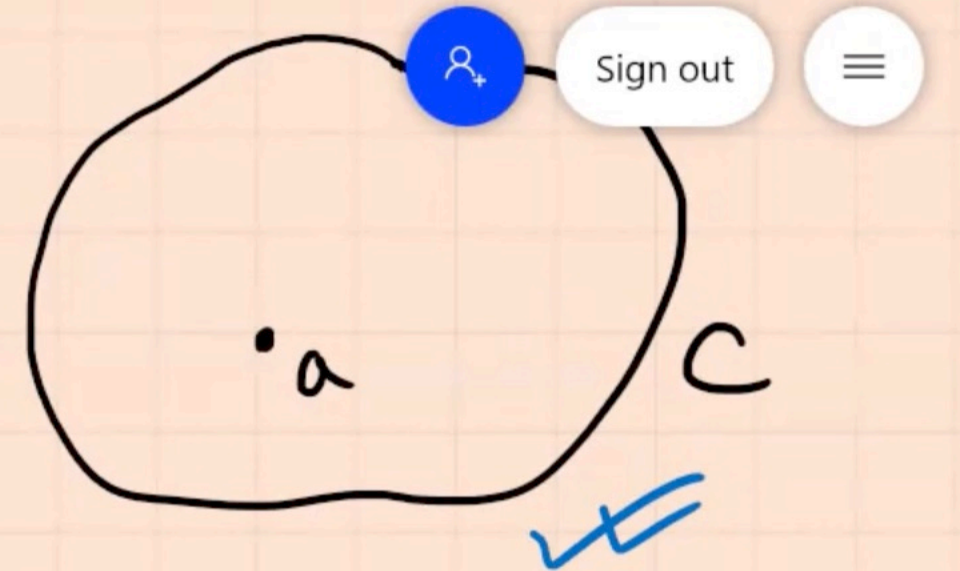
within C , then

$$\int_C \frac{f(z) dz}{z-a} = 2\pi i f(a)$$

Evaluate:

1. $\int \frac{2\tilde{z} + 3z + 3}{z-1}$, where C is $|z|=2$

$$z-a=0 \\ z=a$$



2. $\int \frac{z^2}{z-2i} dz$, where C is $|z|=1$

$$|z-4|=5$$

Centre $z-4=0 \therefore z=4$
radius, $r=5$

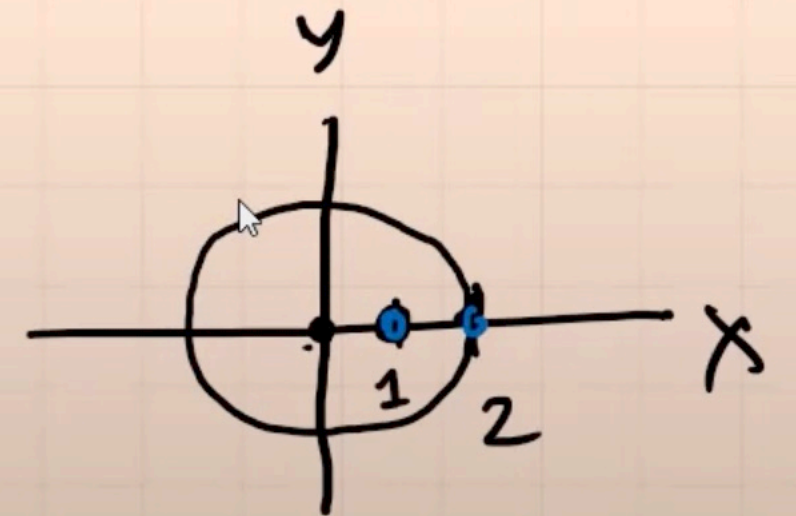
Solⁿ: Pole, $z-1=0 \therefore z=1$ $z=0$ & $r=2$

Youtube search: Cauchy's integral formula problem | Cauchy integral formula in Bangla | Complex Analysis

INSIDE হলে ফর্মুলা
OUTSIDE হলে THEORY

By Cauchy's integral formula

$$\int \frac{2\tilde{z} + 3z + 3}{z-1} = 2\pi i f(1)$$



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$$c \pm d(b)$$

$$\delta \text{ ————— }$$

ANSWEWR: EXAMPLE-182

Here $f(z) = e^z$

$$f(z) = e^z$$

FORMULA

$$f(a) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z-a} dz$$

or $\frac{1}{2\pi i} \oint_C \frac{e^z}{z-2} dz = f(2) = e^2$
[a = 2]

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show that

$$\oint_C \frac{\sin 3z}{z + \frac{\pi}{2}} dz = 2\pi i$$

where C is

circle $|z| = 5$

radius

এইটা দ্বারা theory নাকি ফর্মুলা
ইউজ হবে সেটা বোঝা যায়

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center, $a \Rightarrow z + (\pi/2) = 0$
 $\Rightarrow z = -(\pi/2)$

center < radius হয় তাহলে inside ---> formula
center > radius হলে outside ----> theorem

Answer:

যেহেতু center < radius
তাই ফর্মুলা ইউজ হবে

$$\begin{aligned} \frac{\sin 3z}{z + (\pi/2)} &= 2\pi i \cdot f(a) \\ &= 2\pi i \cdot f(-\pi/2) \\ &= 2\pi i \cdot 1 \\ &= 2\pi i \quad (\text{proved}) \end{aligned}$$

$$\begin{aligned} f(z) &= \sin 3z \\ f(-\pi/2) &= \sin 3(-\pi/2) \\ f(-\pi/2) &= 1 \end{aligned}$$

AE3

Anup Sir

27-01-25

conditional
concept of utility

Math

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Cauchy Integral formula for the 1st derivative.

statement: let $f(z)$ be analytic inside and on a simple closed curve and a is any point inside c .

Then
$$f'(a) = \frac{1}{2\pi i} \int_c \frac{f(z)}{(z-a)^2} dz.$$

$$f''(a) = \frac{2!}{2\pi i} \int_c \frac{f(z)}{(z-a)^3} dz$$

$$f^{(n)}(a) = \frac{n!}{2\pi i} \int_c \frac{f(z)}{(z-a)^{n+1}} dz \quad n = 1, 2, 3, \dots$$

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show that
$$\frac{1}{2\pi i} \int_c \frac{ze^{t_2}}{(z+1)^3} dz = \left(t - \frac{t^2}{2}\right) e^{-t}$$

where c is any simple closed curve enclosing $z = -1$ and $t > 0$

Q.1

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Here $f(z) = ze^{t^2}$

$$\frac{1}{2\pi i} \oint_C \frac{f(z)}{(z-a)^{n+1}} dz = \frac{1}{n!} \left[\frac{d^n}{dz^n} f(z) \right]_{z=a}$$

at $z=a$

for $n=2$, we have

$$\frac{1}{2\pi i} \oint_C \frac{ze^{t^2}}{(z+1)^3} dz = \frac{1}{2!} \left[\frac{d^2}{dz^2} ze^{t^2} \right]_{z=-1}$$

$$\begin{aligned} \frac{1}{2\pi i} \oint_C \frac{ze^{t^2}}{(z+1)^3} dz &= \frac{1}{2} \left[\frac{d}{dz} (e^{t^2} + te^{t^2}) \right]_{z=-1} \\ &= \frac{1}{2} \left[te^{t^2} + t(e^{t^2} + te^{t^2}) \right]_{z=-1} \end{aligned}$$

$$= \frac{1}{2} [2te^{t^2} + t^2e^{t^2}]_{z=-1}$$

$$= \frac{1}{2} [2te^{-t} - t^2e^{-t}]$$

$$= te^{-t} - \frac{1}{2}t^2e^{-t}$$

$$= \left(t - \frac{t^2}{2} \right) e^{-t}$$

(shown)

(pfo)



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$$** \oint_C \frac{f(z)}{(z-a)^{n+1}} dz = \frac{2\pi i}{n!} \left[\frac{d^n}{dz^n} f(z) \right] \text{ at } z=a$$

Here $f(z) = e^{2z}$

for $n=3$

$$\oint_C \frac{e^{2z}}{(z+1)^4} dz = \frac{2\pi i}{3!} \left[\frac{d^3}{dz^3} e^{2z} \right] = \frac{n!}{2\pi i} \int_C \frac{f(z)}{(z-a)^{n+1}} dz$$

$$= \frac{2\pi i}{6} \left[\frac{d^3}{dz^3} e^{2z} \right] \text{ at } z=-1$$

$$= \frac{8\pi i}{6} e^{-2}$$

$$= \frac{2\pi i}{6}$$

(incomplete)

Math

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10-02-25

Laurent theorem

*** Expand the function $f(z) = \frac{1}{z-3}$ is a Laurent series for the (i) $|z| < 3$.

(ii) $|z| > 3$

(i) we have $|z| < 3 \Rightarrow 3 > |z|$ and using this condition, we have

$$\begin{aligned} f(z) &= \frac{1}{z-3} = \frac{1}{-3} \left(1 - \frac{z}{3}\right)^{-1} \quad \text{সূত্র আছে...} \\ &= -\frac{1}{3} \left(1 + \frac{z}{3} + \frac{z^2}{9} + \frac{z^3}{27} + \frac{z^4}{81} + \dots\right) \\ &= -\frac{1}{3} \left(-\frac{z}{9} - \frac{z^2}{27} - \frac{z^3}{81} - \frac{z^4}{243} - \dots\right) \end{aligned}$$

$|z| > 3$

$$f(z) = \frac{1}{z-3} = \frac{1}{z} \left(1 - \frac{3}{z}\right)^{-1}$$

*** Expand $f(z) = \frac{1}{z(z-2)}$ is a -

(i) $0 < |z| < 2$ (ii) $|z| > 2$

$$\begin{aligned} \frac{1}{z(z-2)} &= \frac{A}{z} + \frac{B}{z-2} \\ 1 &= A(z-2) + Bz \\ A &= -\frac{1}{2} \end{aligned}$$

$$f(z) = -\frac{1}{2z} + \frac{1}{2(z-2)}$$

$$= \frac{1}{2} \left[\frac{1}{z-2} - \frac{1}{z} \right]$$

$$f(z) = \frac{1}{2} \left[-\frac{1}{2} \left(1 - \frac{z}{2} \right)^{-1} - \frac{1}{z} \right]^{-1/2}$$

$$(1-x)^{-1} = 1 + x + x^2 + x^3 + x^4 + \dots$$

Expand $f(z) = \frac{z}{(z+1)(z+2)}$

$$1 < |z| < 2$$

$$f(z) = \frac{z}{(z+1)(z+2)} = \frac{z}{z+2} - \frac{1}{z+1} \quad \left\{ \begin{array}{l} \text{Ans for} \\ \text{Q. 10} \\ \text{Ans} \end{array} \right.$$

$$\left. \begin{array}{l} |z| > 1 \\ |z| < 2 \end{array} \right\} \text{Ans}$$

Residue and Residue Theorem

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24-02-25

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Residue at a multiple point

$$a_{-1} = \lim_{z \rightarrow a} \frac{1}{(m-1)!} \frac{d^{m-1}}{dz^{m-1}} \left\{ (z-a)^m f(z) \right\}$$

1	0	0	0
0	1	1	0
1	0	0	1

$z = a$
 $(z-a)^m$ top order

Evaluate the residues of $f(z) = \frac{z^n}{z^n + a^n}$ at the poles

The poles of $f(z)$ are given

by $z^n + a^n = 0$

are simple pole,

$z = \pm ai$ and they

Now $\text{res}(ai) = \lim_{z \rightarrow ai} \left\{ (z-ai) \frac{z^n}{z^n + a^n} \right\}$

$$= \lim_{z \rightarrow ai} \frac{(z-ai) z^n}{(z+ai)(z-ai)}$$

$$= \lim_{z \rightarrow ai} \frac{z^n}{z+ai} = \frac{(ai)^n}{2ai}$$

$$= \frac{a^n i^n}{2ai}$$

$$= \frac{1}{2} ai$$



$$f(z) = \frac{p(z)}{q(z)} \quad z = \pm ai$$

$$a_{-1} = \frac{p(a)}{q'(a)}$$

$$= \frac{(ai)^r}{(r+1)}$$

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$$z = ai$$

$$p(z) = z^r$$

$$q(z) = z^r + a^r$$

$$q'(z) = rz$$

page

$$f(z) = \frac{z - rz}{(z+1)^r (z^r + a^r)}$$

$$(z+1)^r (z^r + a^r) = 0$$

$$\Rightarrow (z+1)^r = 0$$

$$z = -1 \text{ is a}$$

$$\Rightarrow z^r + a^r = 0$$

$$z = \pm ai$$

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Residue Theorem: If $f(z)$ is analytic inside and on a simple closed curve C except at a finite number of singular points $a_1, a_2, a_3, \dots, a_n$ inside C . Then

$$\oint_C f(z) dz = 2\pi i \left[\text{Res}(a_1) + \text{Res}(a_2) + \dots + \text{Res}(a_n) \right]$$

32(37) 50
Analytic
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Chapter - 6

Math

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Contour integration we will evaluate a variety type of 'real definite' integrals with the help of the Cauchy's residue theorem using a suitable type of closed path on this reason the process is called contour integration.

$$\int_0^{\infty} \frac{dx}{x^r+1}$$

$$\int_0^{\pi} \frac{d\theta}{\sin\theta + \cos\theta + 1}$$

$$\textcircled{1} \int_0^{2\pi} f(\cos\theta, \sin\theta) d\theta$$

$$\text{let } I = \int_0^{2\pi} f(\cos\theta, \sin\theta) d\theta \quad \text{--- } \textcircled{1} \text{ where}$$

$f(\cos\theta, \sin\theta)$ is a rational function

of $\cos\theta$ and $\sin\theta$.

$$\text{let } z = e^{i\theta} \text{ then } \cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$$

$$\sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

$$\frac{z + \frac{1}{z}}{2}$$

$$= \frac{z^r + 1}{2z}$$



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$$I = \oint_C f(z) dz$$

where C is the unit circle $|z| = 1$

Evaluate $\int_0^{2\pi} \frac{d\theta}{3 - 2\cos\theta + \sin\theta}$ by the contour integration

$$I = \oint_C \frac{dz}{iz \left(3 - \frac{z^2+1}{2} + \frac{z^2-1}{2i} \right)}$$

$$\sin\theta = \frac{z^2+1}{2iz}$$

$$\cos\theta = \frac{z^2+1}{2z}$$

$$= \oint_C \frac{dz}{iz \left(6iz - 2iz^2 - 2i + z^2 - 1 \right)}$$

$$\Rightarrow \oint_C \frac{2dz}{(1-2i)z^2 + 6iz - 1-2i}$$

$$\Rightarrow (1-2i)z^2 + 6iz - 1-2i = 0$$

$$\Rightarrow z = -6i$$

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Math

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2-03-25

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Math

Radical Sir

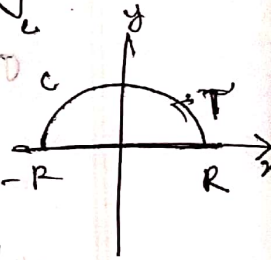
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Type 3:

$$\int_{-a}^a f(x) \cos mx \, dx \quad \text{or} \quad \int_{-a}^a f(x) \sin mx \, dx \quad \text{where}$$

$f(x)$ is a rational function of x ; consider $\oint_C e^{imz} f(z) \, dz$ where C is the contour consisting of



- (i) The x -axis from $-R$ to R where R is larger
- (ii) the upper semi-circle T of the circle

$|z| = R$ which lies above the x -axis

Then let $R \rightarrow \infty$ and if $f(z)$ is an even function this can be used to evaluate the integral. This

$$\lim_{R \rightarrow \infty} \int_T e^{iz} f(z) \, dz = 2\pi i \sum \text{Residues}$$

show that $\int_0^a \frac{\cos mx}{x^2+a^2} dx = \frac{\pi}{2a} e^{-ma}$.

Solⁿ:

consider $\oint_C f(z) dz$ where $f(z) = \frac{e^{imz}}{z^2+a^2}$

The poles of $f(z) = \frac{e^{imz}}{z^2+a^2}$ are obtained by solving $z^2+a^2=0 \Rightarrow z = \pm ai$ and

they are simple poles of which only $z=ai$ lies inside

$$\text{Res}[f(z), ai] = \lim_{z \rightarrow ai} (z - ai) \left\{ \frac{e^{imz}}{(z - ai)(z + ai)} \right\} = \frac{e^{-ma}}{2ai}$$

Now by the Cauchy residue theorem we have

$$\oint_C f(z) dz = 2\pi i \text{Res}[f(z), ai]$$

$$\text{or, } \int_{-R}^R f(x) dx + \int_T f(z) dz = 2\pi i \left(\frac{e^{-ma}}{2ai} \right)$$

$$= \frac{\pi e^{-ma}}{a} \quad \text{--- (1)}$$

Now by Jordan's Lemma,

$$\lim_{R \rightarrow \infty} \int_T f(z) dz = 0 \quad \text{--- (2)}$$

Now taking limit $R \rightarrow \infty$ in (1) and using

eqn (2) $\Rightarrow$

$$\int_{-a}^a f(x) dx = \frac{\pi e^{-ma}}{a} = 2 \int_0^a f(x) dx = \frac{\pi e^{-ma}}{a}$$

$$\text{or, } \int_0^a \frac{\cos mx}{x^2+a^2} dx = \frac{\pi e^{-ma}}{2a}$$

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$$\int_0^{\infty} \frac{e^{-ax} \cos x}{1+x^2} dx \quad \int_0^{\infty} \frac{e^{-ax} \cos mx}{x^2+1} dx \quad \int_0^{\infty} \frac{e^{-ax} \cos x}{x^2+a^2} dx$$